

Soft Copy Assignment: NumPy and Pandas on SMS SPAM COLLECTION

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Batch: SE2(S24)
Dataset Name: SMS SPAM COLLECTION

20 Problem Statements with Solutions

1. Display first 5 rows

```
df.head()
```

2. Display last 5 rows

```
df.tail()
```

3. Find number of rows and columns

```
df.shape
```

4. Show column names

```
df.columns
```

5. Check null values

```
df.isnull().sum()
```

6. Count Spam and Ham messages

```
df['v1'].value_counts()
```

7. Find percentage of Spam messages

```
df['v1'].value_counts(normalize=True)*100
```

8. Find duplicate rows

```
df.duplicated().sum()
```

9. Remove duplicate rows

```
df.drop_duplicates(inplace=True)
```

10. Add message length column

```
df['length']=df['v2'].apply(len)
```

11. Find average message length

```
np.mean(df['length'])
```

12. Find maximum message length

```
np.max(df['length'])
```

13. Find minimum message length

```
np.min(df['length'])
```

14. Show longest message

```
df[df['length']==df['length'].max()]
```

15. Show shortest message

```
df[df['length']==df['length'].min()]
```

16. Average length of Spam and Ham messages

```
df.groupby('v1')['length'].mean()
```

17. Count messages containing word Free

```
df[df['v2'].str.contains('Free', case=False)].shape[0]
```

18. Count messages containing word Win

```
df[df['v2'].str.contains('Win', case=False)].shape[0]
```

19. Sort messages by length

```
df.sort_values(by='length', ascending=False)
```

20. Random 5 messages from dataset

```
df.sample(5)
```

Conclusion: Using NumPy and Pandas, the SMS spam dataset was analyzed successfully.